

# Jayesh Singh

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## SKILLS SUMMARY

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- **Programming Languages:** C++, C#, C, Scripting
- **Engines & APIs:** Unreal Engine 5, Unreal Engine 4, Unity Engine, OpenGL
- **Tools & Technologies:** Git, Perforce, Visual Studio, VR (Oculus SDK, OpenXR), AR (ARCore, Geospatial API)
- **Concepts:** Object-Oriented Programming, Prototyping, Gameplay Systems Design, AI Programming, Performance Optimization, UI/UX Implementation

## PROJECTS

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- **Project: Brawler** (Unreal Engine 5, In Progress):
  - **Combat System:** Architected a clean, data-driven C++ melee combat system supporting multi-stage combos, handling input buffering and to ensure responsive attacks.
  - **Enemy AI & Behaviors:** Engineered custom AI Controllers using C++ and Unreal's Behavior Trees, utilizing Blackboards to create dynamic decision-making logic for multiple enemy types.
  - **Locomotion:** Engineered and prototyped a versatile 3D character controller in C++ with locomotion features including walking, sprinting, crouching, and dodging, integrated with a **Blueprint Animation State Machine** for fluid movement.
  - **Dynamic Animation:** Implemented dynamic gameplay animations by triggering **Animation Montages** from C++ for evasive dodges and context-sensitive enemy hit reactions, providing clear and impactful combat feedback.
- **Shards of Kalia:** Worked as the **AI programmer** on a **2D local coop** game with magic abilities where I designed and implemented enemy behaviors using **State Machines** with some work on UI and sound effects too by **communicating and cross-discipline collaboration**. (Unity)
- **Drag Race:** Built a **split screen local multiplayer** demo in which players get to drive a car, there are oncoming vehicles and obstacles on the road and the camera view can be changed from 3rd person view to driver view. Used **physics** to make the car accelerate and decelerate accurately. (Unity)
- **Feed the animals:** Built a top-down view game with animals coming towards the player from the top and shoot steak to feed them. Used **Object Pooling** for large number of objects for optimization & **increase the performance by 30%**. (Unity)

## EXPERIENCE

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- **BitForwardGames** Rochester, NY  
*Network Programmer & Gameplay Programmer* Sep 2025 - Present
  - **Spell Code SlingerZ** (Available for wishlist on Steam)
    - \* **Network Architecture:** Engineered a peer-to-peer networking architecture, implementing custom rollback netcode to ensure synchronized, low-latency gameplay in a fast-paced multiplayer environment.
    - \* **Systems Integration:** Integrated Facepunch.Steamworks for seamless platform connectivity and matchmaking.
    - \* **Collaboration:** Managed version control and task delegation across disciplines, ensuring stable builds and smooth feature integration for the team.
- **A3ED** Remote  
*Unity Developer Freelancer* Mar 2023 - Jan 2024
  - **VR Training Simulation:** Engineered an interactive VR training simulation in Unity with the Oculus SDK, enabling users to manipulate a complex bulldozer model with 5 interactable components. The simulation allowed for detailed inspection and disassembly, designed to improve user comprehension of complex machinery.
  - **AR Animals:** Created an AR project in Unity powered by Google's ARCore. The users would be able to see animals in AR at selected areas chosen in real world locations with 3D maps with the help of Cesium combined with newly introduced Geospatial from Google.
- **FanClash11 Fantasy Private Limited** Remote  
*Junior Intern Developer* Oct 2021 - Nov 2021
  - **Documentation:** Authored a technical design brief for a new mobile title, outlining core mechanics, engine requirements, and performance targets to align the development team.
  - **UI/UX Prototyping:** Prototyped a responsive UI and menu system using Unity Scene Management, optimizing asset loading to achieve level transitions under **10 seconds** on target mobile devices.
  - **Gameplay Programming:** Implemented core C# gameplay mechanics for the player character, **profiling and optimizing scripts** to guarantee a consistent **30 FPS** on mobile hardware.

## EDUCATION

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- **Rochester Institute of Technology** Rochester, New York  
*MS - Game Design and Development ; GPA: 4.0* 2024 - 2026  
*Courses: Game Dev Processes, Game Design, AI Programming, Game Graphics Programming, Game Engine Development*
- **Bharati Vidyapeeth's College Of Engineering, GGSIPU** Delhi, India  
*Bachelor of Technology - Information Technology; GPA: 8.37* 2018 - 2022